

Day 18 - Lecture 18.10: Adding Labels to Bars (`plt.text`)

1. Overview & Purpose

While bar heights give a visual comparison, viewers frequently need the **exact numerical value** without having to trace their eyes horizontally to the Y-axis. Adding direct value labels on top of each bar drastically improves readability.

2. Syntax & Mechanics: `plt.text()`

```
plt.text(  
    x,                # X coordinate (center of the bar)  
    y,                # Y coordinate (top of bar + offset)  
    s,                # String text to display  
    ha='center',      # Horizontal alignment: 'center', 'left', 'right'  
    va='bottom',      # Vertical alignment: 'bottom', 'top', 'center'  
    fontsize=10,       # Text font size  
    fontweight='bold'  # Font weight  
)
```

The Problem of Top Clipping & `plt.ylim()`:

If a bar has height 1005 and your Y-axis tops out at 1005, putting text at `y = 1005 + 20` will push the text outside the canvas boundary! **Solution:** Always expand the upper Y-limit using:

```
plt.ylim(0, max(revenue) + 200)
```

3. Code Implementation

```
import matplotlib.pyplot as plt

years = [2008, 2009, 2010, 2011, 2012]
revenue = [1005, 170, 427, 133, 232] # in $M

plt.figure(figsize=(8, 5))
plt.bar(years, revenue, color="purple", width=0.55, edgecolor="black")

# Add numerical values above each bar
for i in range(len(revenue)):
    plt.text(
        x=years[i],
        y=revenue[i] + 25,          # Offset above the bar top
        s=f"${revenue[i]}M",        # Formatted string
        ha="center",                # Horizontally centered
        fontweight="bold"
    )

# Add top padding so the highest label is not clipped
plt.ylim(0, max(revenue) + 180)

plt.xlabel("Years", fontsize=12)
plt.ylabel("Revenue (in $M)", fontsize=12)
plt.title("Yearly Revenue of Movies with Value Labels", fontsize=14, fontweight='bold')
plt.grid(axis='y', linestyle=':', alpha=0.6)
plt.show()
```

4. Key Takeaways

- Always pair direct data labels with `plt.ylim()` to guarantee adequate headroom.
- In modern Matplotlib (v3.4+), `plt.bar_label()` is also available as an automatic alternative.